Xi (Xavier) Zhang

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Education

The University of Queensland Feb 2026 – Dec 2026
Bachelor of Economics (Honours) with Honours Class I
Supervisors: Allan Hernandez Chanto and Claudio Mezzetti.
GPA: 7.0/7.0.

The University of Queensland Jul 2024 – Dec 2025
Bachelor of Economics
Minor: Mathematics.
GPA: 7.0/7.0.

Shandong University Sep 2022 – Jun 2024
Bachelor of Economics (Finance) — *awarded June 2026*
Track: Financial Mathematics.
GPA: 93.2/100 (WES-evaluated: 4.0/4.0); Major Rank: 1/14.

Standardized Tests

GRE: Quantitative 170, Verbal 158.

Research Interests

Microeconomic theory, mechanism design, auction theory, and game theory.

Research Experience

Collective Purchasing under Private Information: Efficiency and Cost Recovery Feb 2026 – Nov 2026
Honours thesis; The University of Queensland

- Study collective purchasing in which a fixed group of users with privately known productivities bids through a mediator against outside bidders in a second-price auction for a divisible resource sold as a single lot.
- Under a regularity condition, characterize a surplus-maximizing mechanism that uses adjusted-productivity weights for bidding and internal allocation while satisfying Bayesian incentive compatibility, interim individual rationality, and expected cost recovery.
- Prove that VCG implements the exact first best with expected cost recovery for all sufficiently large finite groups with i.i.d. productivities and fixed total capacity.

The Power of the Pool: Collective Bidding for Scarce Capacity Oct 2025 – Present
Supervisors: Allan Hernandez Chanto and Claudio Mezzetti. Work in progress.

- Study decentralized team bidding in a second-price auction, where privately informed members contribute to a joint bid and a post-award VCG mechanism allocates scarce capacity.
- Prove existence of a symmetric monotone Bayesian Nash equilibrium, characterize equilibrium contributions under a uniform outside-bid benchmark, and analyze payment sharing, endogenous entry, and efficiency.

Winter Research Program, The University of Queensland Jul 2026
Supervisor: Allan Hernandez Chanto. Project: *Group Security Auctions.*

- Developed proofs for a model of teams competing in first-price security auctions, including the separating-equilibrium characterization and its closed-form solution.
- Proved incentive compatibility and individual rationality of the internal mechanism implementing the integrated-team equilibrium.

Summer Research Program, The University of Queensland Nov 2024 – Feb 2025
Supervisor: Yaying Zhou. Project: *The Effect of Parental Retirement on Marriage Rate.*

- Compared five Chinese household surveys for a project on parental retirement and children's marriage, assessing coverage and availability of relevant variables.
- Conducted empirical analysis using STATA and R.

Selected Coursework

Mathematics and Statistics

The University of Queensland (all 7/7): Functional Analysis (including point-set topology; MATH3402), Mathematical Analysis (real analysis; MATH2400), Optimisation Theory (MATH3404), Numerical Linear Algebra & Optimisation (MATH3204), Probability Models & Stochastic Processes (STAT3004), Calculus & Linear Algebra II (MATH2001), Multivariate Calculus & ODEs (MATH1052).

Shandong University: Linear Algebra (98/100), Probability and Mathematical Statistics (94/100), Advanced Mathematics I (95/100), Advanced Mathematics II (90/100).

Economics

The University of Queensland (all 7/7): Microeconomics for Honours (ECON6101), Macroeconomics for Honours (ECON6102), Econometrics for Honours (ECON6103), Research Applications in Microeconomics (ECON6201), Advanced Microeconomics (ECON3010), Advanced Macroeconomics (ECON3020), Econometric Analysis (ECON3330), Micro Econometrics (ECON3360).

Shandong University: Intermediate Microeconomics (94/100), Intermediate Macroeconomics (95/100), Econometrics I (96/100), Game Theory (97/100).

Teaching Experience

Teaching Assistant, The University of Queensland 2026
ECON1050: Tools of Economic Analysis (Semesters 1 and 2, 2026).
MATH1051: Calculus & Linear Algebra I (Semesters 1 and 2, 2026).
ECON2050: Mathematical Economics (Semester 2, 2026).

- Lead tutorials, hold consultations, and grade assessments.
- Develop original tutorial handouts and share them on my academic website.

Teaching Assistant, Shandong University Sep 2023 – Jun 2024
Introductory Microeconomics and Introductory Macroeconomics.

Honours and Scholarships

The University of Queensland

Archibald Scholarship	2026
Winter Research Scholarship	2026
Dean's Honour Roll, Faculty of Business, Economics and Law	2025
Summer Research Scholarship	2024
Dean's Commendation for Academic Excellence (all semesters)	2024–2026

Shandong University

Outstanding Graduate Award (university level)	2026
Outstanding Student Award, First Class	2023
Merit Student	2023

Miscellaneous

Programming: Python, R, Stata, MATLAB, and LaTeX.

Authored LaTeX notes for economics and mathematics courses, available in the Notes on my academic website.

Languages: English (fluent), Chinese (native).

References

Allan Hernandez Chanto
Senior Lecturer
The University of Queensland
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Claudio Mezzetti
Professor (Colin Clark Chair)
The University of Queensland
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Kun Zhang
Assistant Professor in Economics
Auburn University; UQ (on leave)
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